

States of Matter

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A **solid** keeps its own shape and volume; its particles are packed tightly. A **liquid** takes the shape of its container but keeps its volume; its particles slide past each other. A **gas** fills its whole container; its particles are far apart. **Heating** adds energy so particles move more; **cooling** slows them. Heating and cooling can change the state (melt, freeze, evaporate, condense).

Round 1 — Retrieve It

1. Which state of matter keeps its own shape and volume?

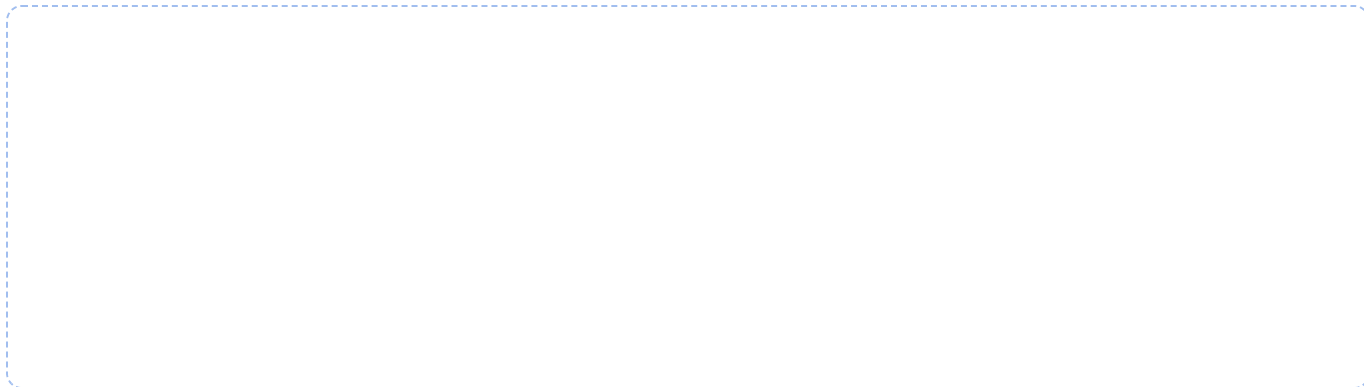
2. What happens to a liquid when you pour it into a new container?

3. How are the particles in a gas different from the particles in a solid?

4. Does heating make particles move faster or slower?

Round 2 — Sketch It

Draw three boxes labeled solid, liquid, and gas. Inside each box, draw dots to show how the particles are arranged in that state.



States of Matter — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. You leave an ice cube on a warm counter. What state change happens, and why?

2. Water in a pot boils and steam rises. Name the state change and tell what the particles are doing.

3. You put liquid water in the freezer overnight. What state change happens, and did you add or remove energy?

Round 4 — Reason from Evidence

A balloon is filled with air and then put in a cold freezer. The balloon gets a little smaller. Use what you know about cooling and particles to explain why.

States of Matter — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how the particles in a solid, a liquid, and a gas are different. Use the words packed, slide, and far apart.

C**Connect**

Connect heating and cooling to something you have seen at home, such as ice melting or fog on a cold window. Name the state change.

E**Extend**

Describe how you could take the same water from ice to liquid to gas and back again. Tell what you would add or remove at each step.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — States of Matter

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A solid.
2. It takes the shape of the new container but keeps the same volume (amount).
3. Gas particles are far apart and spread out to fill the container; solid particles are packed tightly together.
4. Faster (heating adds energy, so particles move more).

Sketch It — look for

Solid: dots packed close together in a fixed pattern. Liquid: dots close but able to slide past each other, filling the bottom. Gas: dots spread far apart, filling the whole box.

Apply It

1. Melting: the ice (solid) gains heat energy and turns into liquid water.
2. Evaporation (boiling): liquid water changes to gas; the particles gain energy and move much faster and farther apart.
3. Freezing: liquid water changes to solid ice; energy is removed (the freezer takes heat away).

Reason from Evidence — look for

Cooling removes energy, so the air particles slow down and move a little closer together, taking up less space. This makes the balloon shrink.

ACE — Extend look-for

Steps show adding heat to go ice → liquid → gas (melt, then evaporate) and removing heat to go gas → liquid → ice (condense, then freeze).

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.